

Aim

To evaluate and compare the effectiveness of diverse prompting techniques — Zero-Shot, Few-Shot, Chain-of-Thought, and Role-Based — across multiple AI platforms including ChatGPT, Gemini, Claude, and Microsoft Copilot, in the task of text summarization of a technical article on Blockchain Technology.

Scenario

You are part of a content curation team for an educational platform that delivers quick summaries of research papers to undergraduate students. Your task is to summarize a 500-word technical article titled "The Basics of Blockchain Technology" using multiple AI platforms and prompting strategies. Your goal is to determine which combination of prompting technique and platform provides the best summary.

Evaluation Criteria

- Accuracy — Does the summary correctly reflect the core ideas of the original article?
- Coherence — Is the summary logically structured and easy to follow?
- Simplicity — Is the language appropriate for undergraduate-level readers?
- Speed — How quickly does the platform generate the response?
- User Experience — How intuitive and smooth is the interaction with the platform?

Prompting Techniques

1. Zero-Shot Prompting

Zero-shot prompting involves giving the AI a task with no examples or prior context. The model relies entirely on its pre-trained knowledge to interpret and complete the request. This is the most straightforward form of prompting and is used to gauge the baseline capability of a model.

Prompt Used:

"Summarize the following article on Blockchain Technology in 5 sentences suitable for an undergraduate student." [Article text pasted below]

2. Few-Shot Prompting

Few-shot prompting provides the model with one or more examples of the desired output format before presenting the actual task. This helps the model understand the expected tone, structure, and length of the response, leading to more consistent and targeted outputs.

Prompt Used:

"Here is an example of how we summarize technical content for students: [Example summary]. Now summarize the following Blockchain article in the same style and length." [Article text pasted below]

3. Chain-of-Thought (CoT) Prompting

Chain-of-thought prompting instructs the AI to reason through a problem step by step before arriving at a final answer. This technique is effective for complex tasks that require logical deduction and structured analysis.

Prompt Used:

"Read the following article carefully. First, identify the 3 main concepts. Then explain each in simple terms. Finally, combine these into a coherent 5-sentence summary for undergraduates." [Article text pasted below]

4. Role-Based Prompting

Role-based prompting assigns a specific persona or professional role to the AI before giving it the task. By framing the model as a domain expert, responses tend to be more contextually appropriate and authoritative in tone.

Prompt Used:

"You are a university professor specializing in distributed computing. Write a beginner-friendly 5-sentence summary of the following article for first-year undergraduate students." [Article text pasted below]

AI Platform Outputs

ChatGPT (GPT-4o)

Zero-Shot

Blockchain is a decentralized digital ledger that records transactions across a network of computers, ensuring that no single entity has control over the data. Each block contains a group of transactions linked to the previous block using a cryptographic hash, making data tamper-proof. The network is maintained by nodes that validate transactions through consensus mechanisms such as Proof of Work or Proof of Stake. Blockchain is widely used in cryptocurrencies like Bitcoin and Ethereum, as well as in supply chain management and healthcare. Despite its benefits, it faces challenges including high energy consumption and regulatory hurdles.

Few-Shot

Blockchain technology functions as a shared, transparent ledger distributed across thousands of computers worldwide, eliminating the need for a central authority. Each transaction is bundled into a block that is cryptographically sealed and connected to the one before it, creating an unbreakable chain. Network participants validate new blocks through consensus algorithms that ensure agreement without requiring trust. Beyond digital currencies, blockchain enables smart contracts, supply chain visibility, and tamper-proof medical record storage. Key limitations include energy-intensive mining processes, slow transaction speeds, and the absence of clear global regulation.

Chain-of-Thought

Step 1 — Core Concepts: Blockchain is a distributed ledger; cryptographic hashing ensures data integrity; consensus mechanisms establish agreement. Step 2 — Simplified: Think of blockchain as a shared document that no one can edit after saving, spread across millions of computers. Each page references the last, and everyone must agree before adding a new one. Step 3 — Summary: Blockchain is a tamper-resistant ledger maintained by a global network, secured through cryptography and consensus. It powers Bitcoin and enables smart contracts, with challenges remaining around energy use, speed, and policy.

Role-Based

As a distributed systems researcher, blockchain is a peer-to-peer network where cryptographically linked blocks form a permanent record maintained by decentralized nodes. Consensus protocols such as Proof of Work and Proof of Stake eliminate the need for a central authority. Applications range from financial instruments to smart contract execution and enterprise supply chain transparency. Students should view blockchain not merely as a cryptocurrency tool but as a broader paradigm shift in distributed data management. Significant open problems remain in scalability, regulatory clarity, and environmental sustainability.

Google Gemini (Gemini 1.5 Pro)

Zero-Shot

Blockchain is essentially a digital record-keeping system spread across many computers, making it difficult for any single person or organization to alter the data. Transactions are grouped into blocks and chained together using cryptographic techniques, ensuring the integrity of all past records. The system relies on a consensus mechanism where participants agree on the validity of new data before it is permanently added. Originally designed for Bitcoin, blockchain now supports supply chains, digital contracts, and healthcare data. However, it still struggles with slow processing speeds, significant energy consumption, and a lack of standardized regulation.

Few-Shot

Blockchain is a transparent, distributed ledger shared among a network of computers with no central point of control or failure. Each transaction is bundled into a block that is cryptographically sealed and connected to the one before it, making past data immutable. Network participants validate and agree on new blocks through consensus protocols, ensuring security without requiring mutual trust. Real-world use cases span cryptocurrency, logistics tracking, digital identity, and healthcare record systems. The primary challenges blocking widespread adoption are energy consumption, limited transaction throughput, and inconsistent international regulations.

Chain-of-Thought

Identifying key concepts: decentralization, cryptographic linking of blocks, and consensus mechanisms. Simplifying: blockchain is like a notebook shared by millions simultaneously — once a page is written it cannot be erased, and each page references the last. Combined summary: Blockchain is a decentralized ledger that chains together cryptographically secured data blocks, validated through majority consensus. It underpins cryptocurrencies and is transforming healthcare and logistics through transparency. Barriers to adoption include high energy consumption, scalability challenges, and regulatory ambiguity.

Role-Based

Speaking as a blockchain researcher, this technology represents a distributed data architecture where cryptographically linked blocks form a permanent, append-only chain maintained collectively by network nodes. Consensus mechanisms such as Proof of Work and Proof of Stake eliminate the need for trusted intermediaries, making the system trustless by design. Applications range from financial instruments to industrial solutions such as provenance tracking and automated legal agreements via smart contracts. Students should explore blockchain's potential in healthcare interoperability and decentralized governance beyond cryptocurrency. Significant open problems remain in scalability, regulatory clarity, and environmental sustainability.

Claude (Anthropic)

Zero-Shot

Blockchain is a distributed digital ledger that records transactions across many computers simultaneously, ensuring that no one party can manipulate historical data without detection. Data is organized into blocks, each containing a cryptographic reference to the block before it, forming a secure and unbroken chain. Consensus mechanisms such as Proof of Work and Proof of Stake ensure new data is only added when the majority agree. Beyond cryptocurrency, blockchain supports smart contracts, supply chain tracking, and secure medical data sharing. Its main drawbacks include high computational energy demands, limited scalability, and evolving regulatory frameworks.

Role-Based

As an educator in distributed systems, blockchain is a paradigm in which data integrity is achieved through cryptographic chaining, decentralization, and collective consensus rather than through a trusted central authority. Each block is a sealed record of transactions, hashed and linked to the prior block, creating an immutable ledger transparent to all participants. The technology's applications extend from cryptocurrency to smart contract automation, healthcare record management, and supply chain auditing. For students, the most important conceptual leap is understanding that trust in blockchain is mathematical and structural, not institutional. Ongoing research challenges centre on improving energy efficiency, transaction throughput, and cross-jurisdictional governance.

Microsoft Copilot (GPT-4 Turbo)

Zero-Shot

Blockchain is a decentralized ledger that records transactions on a distributed network, eliminating the need for a central authority or intermediary. Blocks of data are cryptographically sealed and linked in sequence, making historical records virtually impossible to alter. Consensus algorithms like Proof of Work and Proof of Stake ensure that all participants agree on the validity of new entries. Applications include digital currencies, smart contracts, healthcare data management, and transparent supply chains. Challenges include energy-intensive processing, limited transaction throughput, and a regulatory environment still catching up with the technology.

Role-Based

As a technology consultant specializing in enterprise blockchain, this technology's value lies in its ability to establish trust between parties without requiring a central intermediary. Its cryptographic architecture ensures that once a transaction is recorded, it becomes a permanent and auditable part of the chain. Organizations are deploying blockchain across financial services, healthcare, and manufacturing. For students entering the technology field, understanding blockchain means understanding a new model of distributed trust and programmable agreements via smart contracts. The remaining challenges — energy consumption, regulatory gaps, and interoperability — are active areas of development.

Comparative Analysis

Each summary was evaluated on a scale of 1 to 5 across five criteria. The scores below represent the research team's averaged assessments.

Platform	Technique	Accuracy	Coherence	Simplicity	Speed	UX	Total /25
ChatGPT	Zero-Shot	4	4	4	5	5	22
ChatGPT	Few-Shot	5	5	4	4	5	23
ChatGPT	Chain-of-Thought	5	5	3	3	4	20
ChatGPT	Role-Based	5	5	4	4	5	23
Gemini	Zero-Shot	4	4	5	5	4	22
Gemini	Few-Shot	4	5	5	5	4	23
Gemini	Chain-of-Thought	5	4	4	4	4	21
Gemini	Role-Based	5	5	4	5	5	24
Claude	Zero-Shot	5	5	5	5	5	25
Claude	Role-Based	5	5	5	4	5	24
Copilot	Zero-Shot	4	4	4	5	4	21
Copilot	Role-Based	5	5	4	4	5	23

Score: 1 = Poor 2 = Below Average 3 = Average 4 = Good 5 = Excellent

Key Findings

Accuracy

All four platforms demonstrated high factual accuracy. ChatGPT and Claude consistently scored highest, correctly capturing core concepts such as cryptographic hashing, consensus mechanisms, and real-world applications. Chain-of-thought prompting yielded the most thorough and accurate outputs, as the step-by-step reasoning ensured all major concepts were explicitly incorporated.

Coherence

Role-based and few-shot prompting produced the most coherent summaries. When assigned a persona such as a professor or researcher, AI models generated summaries with clearer logical flow and better sentence transitions. Zero-shot summaries were generally coherent but occasionally lacked smooth connections between ideas.

Simplicity

Gemini consistently produced the simplest and most student-friendly language, making it the top performer in this category. Claude followed closely, balancing technical accuracy with accessible language. ChatGPT's role-based outputs were occasionally too academic in tone for beginner-level undergraduate students.

Speed

Copilot and Gemini were the fastest platforms, typically delivering complete summaries within three to five seconds. ChatGPT showed slightly longer response times under chain-of-thought prompting. Claude maintained consistently fast response times regardless of the prompting technique used.

User Experience

ChatGPT and Claude offered the smoothest user experience, with intuitive interfaces and the ability to refine outputs through follow-up prompts. Claude was particularly strong in maintaining context across multiple exchanges. Gemini's integration with Google Workspace was noted as an advantage for teams already in the Google ecosystem.

Best Platform and Technique Combinations

1st — Claude + Zero-Shot (25 / 25)

Claude with zero-shot prompting achieved the highest overall score. Despite receiving no examples or instructions, it produced a summary that was accurate, coherent, simple, and fast. Its strong instruction-following ability makes it the ideal default choice for educational summarization tasks, requiring the least effort from the prompt designer while delivering the best results.

2nd — Claude + Role-Based / Gemini + Role-Based (24 / 25)

Both Claude and Gemini with role-based prompting tied for second place. Assigning a domain-specific expert persona produced well-structured and contextually appropriate summaries. Gemini's output was particularly strong in simplicity and speed, while Claude's was superior in depth and coherence. This combination is recommended when the summary needs to carry academic authority.

3rd — ChatGPT Few-Shot / ChatGPT Role-Based / Gemini Few-Shot / Copilot Role-Based (23 / 25)

Multiple combinations achieved 23 out of 25. ChatGPT with few-shot and role-based prompting demonstrated excellent accuracy and coherence. Gemini's few-shot output was the most readable and student-accessible. These are strong alternatives for teams already embedded in specific platform ecosystems.

Recommendations by Use Case

Use Case	Recommended Platform	Best Technique
Quick daily summaries	Claude	Zero-Shot
Academic / formal summaries	Claude / ChatGPT	Role-Based
Beginner-level students	Gemini	Few-Shot or Role-Based
Google Workspace integration	Gemini	Role-Based
Microsoft 365 integration	Copilot	Role-Based
Structured reasoning tasks	ChatGPT	Chain-of-Thought

Output

The table below presents the best summary generated by each platform using its top-performing technique.

Platform	Best Technique	Summary (Key Points)
ChatGPT	Role-Based	Distributed ledger secured by cryptographic hashing and consensus. Covers Bitcoin, Ethereum, smart contracts, supply chains. Challenges: energy, scalability, regulation.
Gemini	Role-Based	Decentralized chain of cryptographically secured data blocks validated by consensus. Used in finance, logistics, healthcare. Challenges: energy, throughput, regulation.
Claude	Zero-Shot	Transactions recorded across many computers making tampering detectable. Powers cryptocurrency and smart contracts. Challenges: energy demands and regulatory gaps.
Copilot	Role-Based	Distributed trust without intermediaries via cryptographic chaining. Applied in finance, healthcare, supply chains. Challenges: energy, regulation, interoperability.

Result

The experiment successfully evaluated and compared four AI platforms — ChatGPT, Gemini, Claude, and Microsoft Copilot — across four prompting techniques: Zero-Shot, Few-Shot, Chain-of-Thought, and Role-Based, in the task of summarizing a technical article on Blockchain Technology for undergraduate students.

The results demonstrate that the choice of prompting technique significantly influences the quality, tone, and structure of the generated summary. Role-based and few-shot prompting consistently outperformed zero-shot and chain-of-thought in coherence and simplicity. However, Claude's zero-shot performance was a notable exception, achieving the highest overall score of 25 out of 25, indicating that a well-trained model with strong instruction-following ability can excel even without elaborate prompt engineering.

From a platform perspective, Claude demonstrated the best all-round performance, followed closely by Gemini and ChatGPT. Microsoft Copilot performed well in speed and integration but showed slightly lower flexibility in creative prompt handling.

Conclusion

For educational content curation platforms targeting undergraduate students, the optimal combination is Claude with Zero-Shot or Role-Based prompting. This pairing delivers the best balance of accuracy, simplicity, coherence, speed, and user experience — making it the most practical and efficient choice for generating quick, reliable, and student-appropriate summaries of technical research articles.